Data Science & GenAI Assessment

Submission Deadline: 22-03-2026 11:59 PM

General Instructions:

- Use only free tools such as Google Colab, Jupyter Notebook, or VS Code.
- Do NOT use paid APIs (OpenAI, Claude, Gemini paid versions).
- Submit a Python notebook (.ipynb) with proper markdown explanations.
- Also submit a short PDF (1–2 pages) explaining your approach and key insights.
- Focus on clarity, logic, and clean modular code rather than complexity.
- Ensure your solution runs without errors.

Assignment 1: Store Sales Analysis & Prediction

Dataset Provided: sales_data.csv

- Perform EDA: Check missing values, plot sales over time, compare promotion vs non-promotion sales, and compare sales by DayOfWeek.
- Create at least one lag feature (lag_1). Rolling average feature is optional.
- Use last 30 days as the test set (No random split allowed).
- Train one model.
- Evaluate model using MAE or RMSE.
- Predict next 7 days sales.
- Provide 5 business insights based on your findings.

Assignment 2: Simple Policy Q&A Bot

Documents Provided: leave_policy.txt, it_policy.txt, travel_policy.txt

- Build a simple Q&A system using provided policy text files.
- The system must retrieve relevant information from documents before answering.
- If answer is not found, return: 'Information not available in policy documents.'
- You may use keyword search or TF-IDF with cosine similarity.
- Display which document was used to generate the answer.
- Do NOT use paid APIs or GPU-based large models.